NAVIGATION / +C THROUGH +C9 AUDIO-LOAD-METER RELEASE

Guide navigation

Select a pad and press **INFO twice** for parameters. **Select + INFO** opens supported modulation. Page numbers refer to the full guide.

Waveform / Stop · 2	Pinch vertically for amplitude or horizontally for time. Two-finger tap: 100% amplitude + stock horizontal view. Hold a Trigger grid pad or the Play button on waveform/parameter pages 2 s to stop [+c7].
Speed / Pitch · 3	**Sample > LFO/Spd > SPEED**: 50-200%. One stretch voice, shared with Smooth. Pitch is unrestricted. Waiting voices need restart or neutral-then-adjust.
Tone · 4	**Pad > Tone**: Fidelity, Drive, Filter 2, Res 2, Slope 1. Select + INFO on Drive, Filter 2 or Res 2 to modulate.
Clip / Slicer · 5-6	**Pos**: Clip timing, Live Flip and Fill every; Slicer Sync, Sync Grid and **Slice Fit > Cut / Tape / Smooth**.
GEN · 7-8	**Pad > Gen** for patterns; **Cnf** for extra controls. Enable **Tools > Mstr > Section > GEN Run > Run On**. Follows the musical clock.
Delay / reverb · 9-10	**FX1 > INFO**: **Purge** clears the old delay tail. More: extra controls. **FX2 > INFO**: reverb; More for detail. Tap selected FX2 again for FX3, then INFO.
Sends / Dry · 11, 19	**Pad > Pos**: FX1/2/3 Send and Dry Level; select + INFO to modulate. Dry 0% is wet only. **Tools > Rec > FX Out** routes returns; FX3 has its own Out.
Modulation · 11-12	KEY, ENV, CHAOS, RAND and NET join native sources. Up to 12 saved routes per pad. Double-tap numeric dials to reset base values while keeping routes.
CV / MIDI · 13, 16	**Tools > Rec > CV In**: Scale, Offset, Quantize, Root. **Tools > MIDI > Monitor**: messages, Freeze, channel filter, Clear.
Master · 14-15	**Tools > Comp**: compressor. **Mstr > Section**: Comp env, Master FX, Dist Ctrl, Sweep Mod. Main and Mstr share the same compressor switch.
Overdub · 17	Empty pad > **Cnf > Dub source**: Manual or Pad 1-16. **Dub tap**: Pre / Post pad / Post+FX. Use **In 1/2**; Length Custom ends at Stop.
Recording / Resam · 18	**Rec Input > Resam** prints the mix. Tap destination on Pads to Stop. To reopen without stopping: select with a left encoder + INFO. Right encoders adjust live Gain. Rec starts a stopped clock and restores its prior state after the take [+c7].
Display / mutes · 2, 22, 24	**Tools > Disp**: Brightness, Flip UI, saver, Sig wake; header Gfx toggles lines; CPU toggles the audio load meter (p. 24). **Mixer** mutes now save with the preset. **Main > RESET** keeps pad and Display settings.
Clock [+c6] · 20-21	**Main > Int BPM**: turn faster for larger changes; hold 3 s for 120.0. **Clock**: Run / Stop / Reset, Run CV, Reset CV, Reset CC/Ch. HOME: main Menu. Back: Tools. Stop holds time, not all audio.
Input FX / Install · 23, 25	**Recorder settings > Input FX pad**: Off (default), Follow or Pad 1-16. Use input Filter/Tone, sends and Dry as an effect box. Install/recover: patch official 2.3.4; put **BITBOXMK2.bin** on the card and power on holding INFO.

WAVEFORM +C5 / DISPLAY +C6 / HOLD FIXES +C7

Waveform, pads and display

Pad gestures and global display controls

Waveform and numeric dials

Press INFO on a loaded pad. Pinch vertically for **25-3200% amplitude**, horizontally for time, or diagonally for both. **[+c5]** Horizontal zoom-out reaches four times farther than stock; **LStr / LEnd** values fit long positions. A still **two-finger tap** restores 100% amplitude and stock horizontal zoom-out. Long samples may still extend beyond the screen. Zoom changes only the view.

Double-tap a numeric dial on a pad or FX grid to restore its base value, such as Pitch to zero. Saved modulation stays intact.

Stop a Trigger pad [+c7 fixes]

A **Trigger** pad plays or retriggers on contact. Hold it on the Pads grid, or hold the **Play button** on its waveform or parameter pages, for **two seconds** to release its voices once. Other pads and tails can continue. GEN, MIDI, CV or a new tap can retrigger it. Moving away or leaving the view cancels the hold.

Tools > Disp [+c6 layout]

Brightness / Flip UI	Now in the left column. Adjust screen brightness or reverse the display orientation.
Scrn saver	Right column. Choose **disabled** (default) or **1, 2, 3, 5, 10, 15, 30 or 60 minutes**. Playback continues asleep. The first touch, button or encoder movement only wakes the screen.
Sig wake	Right column: disabled, MIDI, CV trig or both. Signals wake and still play normally. MIDI Clock and Active Sensing are ignored; CV uses rising triggers/gates.
Gfx Off / Gfx On	Header toggle. Off: black screen. On: ten blue, purple and white lines at the current brightness; one changes every half-second.

Renumber the counter [+c3]

Double-tap the top-left bar:beat counter. A star appears; the next bar reads **1:1** for a new session display origin. Playback, clock phase and fills stay unchanged. For real Clock Reset, see page 20.

All Display preferences save globally and survive Tools Reset, including **CPU On/Off [+c9]** in the header (page 24). The stock **MIDI In** tab is labelled **MIDI**.

PITCH COEXISTENCE +C4 / RENDERER AND LOOP FIX +C5

Speed and native Pitch

Sample parameters > LFO/Spd > SPEED

Set duration and pitch independently

Drag SPEED or turn its encoder. **50%** plays the Sample in twice the time and **200%** plays it in half the time. Speed preserves the tuning set by Pitch. Neither control follows BPM directly. At 100% with zero Speed modulation, the voice uses native playback. Select SPEED and press INFO to assign modulation.

Native **Pitch works on every playing pad, including the Sample voice using Speed**. For example, set a 48 kHz Sample to +24 semitones, then compare 50%, 100% and 200% Speed. The rhythm changes while the sample stays two octaves up, within the native Pitch range. Higher file rates and MIDI transposition count toward the original pitch ceiling.

One active stretch voice

First request	Sample Speed and Slicer Smooth share one slot. The first eligible request gets it. A live adjustment on a playing Sample can acquire a free slot. Polyphonic notes on the same pad also share that single slot.
SPDBUSY	A waiting voice plays at its own Pitch and 100% Speed. You can still edit its saved Speed and routes.
RETRY	Once the owner stops or releases the slot, a waiting voice continues at 100%. Restart it, or return Speed to 100%, let that take effect, then adjust again.
Release live	Set the owner to 100% and its Speed modulation depths to zero. The slot becomes available after the short handoff to native playback. A nonzero Speed route holds the slot through crossings of 100%.

Speed and its routes save with the pad. Older presets start at 100%. Reverse playback, loop crossfade, modulated loop boundaries and unsupported source configurations use the native playback path.

Improved in +c5: more efficient rendering reduces copying and interpolation overhead. Supported forward-looped Samples now stretch before LoopStart and continue into the loop. The single-stretch limit remains; native Pitch is unrestricted.

TONE +C2 / CODEC FIDELITY MODES +C3

Tone and Fidelity

Pad parameters > Tone, on Sample, Clip, Slicer and Granular pads.

Fidelity replaces Interp [+c2]

Normal and **HighQ** retain the native choices. **SP1200, SP-12, S950, MPC60 and SP Raw** add sampler-inspired coloration. Fixed-rate choices combine 12-bit quantization with **16, 12, 10, 8, 6, 4 or 2 kHz** rate reduction.

Phone [+c3]	8 kHz mu-law coloration with telephone-band low-pass filtering.
ADPCM [+c3]	IMA-style adaptive 4-bit quantization at 22.05 kHz.
u-Law [+c3]	Mu-law coloration at 16 kHz. The module uses the ASCII label u-Law.

The new modes follow 2k, preserving earlier saved selection numbers. Stereo channels have independent codec histories. Fidelity remains a manual selector.

Drive and two filters [+c2]

Drive	0-100% soft saturation before the filters; default 0%. Select + INFO for modulation.
Slope 1	12 or 24 dB/octave for the original Main-page Filter; default 12 dB. Move Filter away from center to hear the difference. Manual selector.
Filter 2 / Res 2	Independent 24 dB/octave filter. Negative selects low-pass, positive high-pass, center bypasses it. Res 2 defaults to 50%. Both support select + INFO modulation.

The left bank has Fidelity and Drive above Filter 2 and Res 2. Slope 1 sits at the top of the right bank. Tap a bank to give its controls the encoders.

Double-tap resets numeric base values while keeping routes. Returning from modulation restores the same Tone dial and encoder bank. The configuration tab is shortened to **Cnf**. Blank pads expose **LFO/Spd** and Speed at 100%. LFO encoders, touch controls and toggles also edit and save normally on empty pads [+c8 fix]; the native LFO signal still needs an active sample voice.

CLIP TOOLS +C2 / CLOCKED ADDITIONS +C3

Clip timing, fills and slices

Pad parameters > Pos. Scroll the list to reach the added controls.

Loop Fade [+c2]	0-50 ms, default off. Dips loop edges toward zero to reduce clicks. Capped at a quarter of a short loop; try 2-5 ms initially.
Shift [+c2]	-100 to +100 ms. Positive starts later in the file; negative brings its tail forward. Wraps at the source boundary.
Play Len [+c2]	Full or 1/4-32 bars in quarter-bar steps. Caps at the available source, rounding to the nearest Sync division with a one-division minimum.
Sync 1/32 [+c2]	Adds a finer timed Sync division. Play Len, Live Flip and Fill every require Quant Size enabled and a timed Sync division; not Slice sync or None.

Live Flip [+c2], Breakcut and periodic fills [+c3]

Beat Rpt plays beats 1,2,2,2; **Half Rpt** plays 1,2,1,2; **Rotate** plays 2,3,4,1. **1/8 Rpt** and **1/16 Rpt** repeat subdivisions in beat four. **Breakcut** adds changing sixteenth-note cuts/repeats while keeping the downbeat.

Fill every: Always / 2 / 4 / 8 / 16 bars. At 4 bars, for example, bars 1-3 play normally and bar 4 uses the selected Live Flip. Fills follow the underlying musical clock, including short loops. Counter renumbering does not change their phase. Off rejoins the current source position.

These Clip effects run independently of Sample Speed and GEN. Set Beat Count to match the loop. Cuts through sustained sound can click.

Slice Sync Grid [+c3]

For a Slicer pad with **Sync On**, choose Free / 1 bar / 1/2 / 1/4 / 1/8 / 1/16 / 1/32. Synchronized slice duration rounds to a whole grid unit, minimum one. Free uses the native synced duration. **Slice Fit** sets how the Slicer plays within that duration. See page 6. Sync Off uses native playback.

Slice Mono fix [+c2]: after switching a polyphonic Sample pad to Slicer, the saved Mono setting is honored. With Trigger, Mono and Play Through On, a new slice cuts the old one even with Quant Size None.

TAPE +C4 / SMOOTH +C5

Slicer Tape and Smooth

Slicer > INFO twice > Pos > Sync On > Slice Fit

Fit slices to the synced duration

Choose a break with clear kick and snare markers. Enable **Sync**, choose **Sync Grid** if needed, then set **Slice Fit**, immediately below Sync Grid. Launch a fresh slice after changing these settings. Sync Off uses native playback.

Cut	Default for new and existing presets. The native reader plays at the normal pitch rate. The synced duration can trim a slice or leave a gap.
Tape [+c4]	Changes playback rate and pitch together to fit from one marker to the next. A 125 ms slice fitted to 250 ms plays an octave lower; fitted to 62.5 ms it plays an octave higher.
Smooth [+c5]	Fits eligible slices while preserving pitch, at roughly 50-200% fitted rates. Uses the same single stretch slot as Sample Speed.
Pitch	Tape derives pitch from the fitted duration. Smooth preserves the configured native Pitch; Cut retains ordinary Slicer Pitch control. Native Pitch on other pads is unrestricted.

Shared slot and fallbacks

Only one Sample Speed or Slicer Smooth voice can stretch at a time. Tape works independently. If Smooth cannot acquire the slot or its slice/rate is unsupported, that slice uses the native Cut path. A new slice launch can try again.

Very short durations that cannot contain the native fades, invalid markers and extreme rates also use Cut. Slice Fit saves with the pad. The grid chooses duration; Slice Fit chooses how audio occupies it.

Compare **Cut, Tape and Smooth** on the same break at a slower tempo. Tape shifts pitch with duration; Smooth separates them. Then try Play Thru or Slice Seq with the same slice markers.

PER-PAD GENERATORS / +C4

Start a GEN pattern

One clocked generator per pad, up to sixteen running together.

Build a kick-and-hat pattern

1. Load a short kick on a Sample pad. Press INFO twice and open **Gen**. Select **Euclid**, Steps 16, Hits 4 and Rate 1/16. Set another pad with a hi-hat to Euclid, Steps 16 and Hits 5.

2. In **Tools > Mstr > Section > GEN Run**, set **Run On**. The patterns follow the Bitbox musical clock, including MIDI or CV clock. Clock Run/Stop/Reset is on Tools > Main (pages 20-21); GEN Run remains a separate setting. Return to the hat and turn Rotate one step to move it against the kick.

3. Lower Prob to skip occasional hits. Gen Vel sets the generated velocity. Notes sound on the scheduled step. Gate pads release at the programmed gate end. Set Gen to Off to silence one generator. GEN Run > Run Off stops scheduling on all pads and releases their gates. Triggered one-shots and effect tails can finish.

Find the controls

Pad > Gen	Sample, Clip, Slicer and Multisample have eight dials: Gen, Hits, Steps, Rotate, Rate, Prob, Var and Gen Vel. Touch a bank to assign its four encoders.
Pad > Cnf	Gen Cond selects a condition, Gen Proc selects a Breakcut procedure, and Resp Pad chooses the Call+Resp answer pad. Granular keeps its Gran tab and places all eleven GEN controls at the end of Cnf.
Tools > Mstr > Section	GEN Run: Run, Fill, Seed. GEN Play: Epoch, Turn, Mutate. GEN Cut: Cut, Rolls, Recut. GEN Learn: Learn, Scale, Range.

Pad GEN settings save with the preset. Global GEN settings save under Tools. Defaults are Run Off and Gen Off on each pad. If Run is already on, loading a preset can start its configured generators.

PERFORMANCE REFERENCE / +C4

GEN algorithms and status

Var has a different musical function for each algorithm.

Euclid	Distributes Hits across Steps. Rotate shifts the pattern. Var above its midpoint introduces soft ghost hits.
Breakcut	Rearranges break or slice segments. Choose Gen Proc under Cnf, then enable Cut under GEN Cut. Var changes how much is rearranged.
Call+Resp	Three call cycles lead to an answer on the fourth. Choose Resp Pad to send the answer to another pad. Var controls response density.
Evolve	Changes the rhythm over successive cycles. Var controls mutation.
Drummap	Pad columns take kick, snare, hat and percussion roles. Rotate and Var move across the drum map.
Random	Hits controls average density. Var spreads velocities. This is a rhythm generator; the RAND modulation source is described on page 12.
Acid	Generates notes using the global Scale and Range. Var influences accents and slides.
Turing	Generates a repeating pitched pattern. Turn Var toward 100 to lock it.
Learn	Select Learn on the pad and turn on Tools > Mstr > GEN Learn > Learn. Tap the pad to teach timing gaps. Var controls adherence. Learned history lasts for the powered session.

Read the status strip



With Run on, select a pad whose generator is enabled. **Tall white bars** show programmed hits. **Short gray bars** show rests. The **yellow playhead** stays above the current step. Conditions and probability can suppress a programmed hit.

Cyan dashes below the pattern identify enabled generators. A dash flashes white on an emitted note and a yellow underline marks the selected pad. Selecting a pad with Gen Off clears the strip. Recording progress and longer native messages take priority.

DELAY +C3-C4 / TAIL PURGE +C8

Delay and FX return routing

FX1 > INFO for delay. Tools > Rec > FX Out for the physical returns.

Choose **Digital** for the original delay, **Tape** for driven repeats with drift and flutter, **BBD** for darker repeats that grow coarser at longer delay times, or **Codec** for accumulating 4-bit ADPCM coloration. Delay, Feedback, Cutoff, Width, Beat Sync, Filter and Ping retain their native controls.

FX1 > INFO > More

Flutter	Adds 0-100% flutter depth on all four types, default 0. Tape also has built-in flutter.
Mod Rate / Depth	0.05-10 Hz wow rate with up to 10 ms of movement. Depth defaults to 0. Raise Depth to hear Rate.
Phase	Stereo modulation offset from 0 to 180 degrees, initially 90.
Drive / Tone	Drive saturates sound entering the delay line. Tone sets repeat bandwidth. Effects accumulate on successive echoes. Start at Drive 0% and Tone fully open.
Loss / Bandwidth	In Codec mode, Drive becomes Loss (0-30% chance of dropping a whole 10 ms frame). Tone becomes Bandwidth (4-16 kHz codec sample rate). Defaults are 0% loss and 16 kHz.
RETURN DB	Wet return gain from -12 to +12 dB, default 0 dB. This changes echo level on its chosen output independently of Feedback.

Purge [+c8] in the header clears the shared delay tail and feedback history on all four types. New input still echoes; FX2/FX3 continue. More and Main switch pages. Settings save with the preset.

Tools > Rec > FX Out

Out1/2	Default. FX1 delay and FX2 reverb return to the main stereo pair.
Dly FX1/2 / Rvb FX1/2	Moves the named effect to the physical FX stereo pair. The other effect stays on main.
FX1/2 / Dly1 Rvb2	FX1/2 moves both effects to the FX pair. Dly1 Rvb2 sends delay to mono FX1 and reverb to mono FX2, folding each stereo return to mono.

FX Out is global. FX3 has its own output setting. Master distortion and Sweep process Out1/2, while compressor routing follows the selected bus.

Reverb and the third send

FX2 > INFO, or select FX3 using the pad-grid toggle.

Reverb types and main controls [+c2]

Plate is the original algorithm and default. **Room, Hall and Chamber** add Size, Diffusion and Width. All types offer Level: 50% is unity, with about -12 to +12 dB available. Decay, Predelay and Damping retain native modulation. Type changes briefly fade and clear the previous tail.

Room / Hall / Chamber > More [+c3]

Bass x / Xover	Bass decay from 0.5x to 2x relative to midrange; center 1x. Crossover approximately 100 Hz-2 kHz.
Mod Rate / Dpth	Internal reverb modulation, approximately 0.1-5 Hz, with adjustable depth.
Early	Early-reflection level.
Hi Cut / Lo Cut	Input high-frequency cutoff approximately 1-14 kHz; low-frequency cutoff approximately 20-500 Hz.
Level / Main	FX2 repeats its output Level here; FX3 uses this square for FX3 Out and keeps Level on its main page. Main returns without clearing the tail.

FX3: a second independent reverb [+c3]

On Pads, tap FX2 once to select it, then tap the selected cell again. Its label becomes **FX 3**. Press INFO for parameters. Another tap on the selected cell returns it to FX2. FX3 has the same four algorithms and detailed controls, with its own settings and tail.

FX3 Out selects Out1/2 / FX1/2 / FX1 / FX2 independently. It is on Plate's main page, or under More for Room/Hall/Chamber. Mono routing averages both wet channels. An enabled compressor processes the destination pair; master distortion and Sweep remain on main only.

FX3 Send is under each pad's **Pos** page and defaults to zero. FX2 and FX3 settings save independently with the preset. The grid toggle changes only which effect is being viewed.

FX3 skips processing while unused and lets a tail run after its last input. For direct-to-fully-wet balance using all three effects, see Dry Level on page 19.

THIRD CUSTOM RELEASE / +C3

Send modulation and KEY

Pad > Pos > FX1 Send / FX2 Send / FX3 Send > select > INFO

Move the sends with modulation

The usual source/amount editor now works for all three pad sends, in every pad mode, including empty pads. LFO, external CV, velocity, KEY, MIDI CC and other native sources can control the send. HOME returns to Pos and the same selected row.

Modulation adds to the manual send level, then clamps to 0-100% before the native send curve. A zero manual send can therefore open under positive modulation. Remove the route to restore the manual level. The original mixer controls set those same underlying levels.

Each send offers three route slots, within the native **12 saved modulation routes per pad total** across all destinations. This limit also includes Pitch, Speed, filters and other assignments.

What happens with polyphony?

A send processes the pad's summed audio. For voice-dependent **LFO, VEL and KEY**, the last active voice in the native list supplies that pad's send modulation; Clip/Slicer uses its last active embedded voice. This is not an independent send envelope for every note and does not promise newest-note priority.

External CV and controller sources are shared independently of that voice selection. When no voice is active, voice-dependent sources contribute zero. Send settings and routes save with the preset; recording retains the send routes too.

KEY exposes native note tracking

Choose **KEY** after CC in the modulation source selector. It uses the engine's existing note-position source, centered on MIDI note 60. Positive or negative amounts can open or close a filter as you move up the keyboard, for example.

KEY is available on supported native destinations and custom Tone, Speed and send destinations. Per-voice parameters retain per-voice tracking; sends use the pad rule above. KEY differs from **PTCH**, the existing pitch-controller source. Older source-selector indices are unchanged.

PER-PAD MODULATION / +C4

Four pad modulation sources

Select a supported pad parameter, press INFO, choose Source and set its Amount.

After KEY, the source list offers **ENV, CHAOS, RAND and NET**. Each of the 16 pads has independent state for all four. Two routes from one source on the same pad share its current value. Use positive or negative Amount to set the direction and depth.

ENV	A single envelope for the pad follows its Attack, Decay, Sustain and Release settings. Each admitted voice start retriggers the attack from the current level. The source runs from zero to one.
CHAOS	A continuously evolving, fixed-rate bipolar signal generated by a Lorenz attractor. Use a small amount on Filter 2 for irregular movement.
RAND	A new bipolar value on each admitted note, held until the next hit. Repeated triggers on a mono pad also advance it. Try RAND on Pan or FX1 Send.
NET	Four coupled nonlinear nodes produce slow, evolving bipolar movement at a fixed rate. Try NET on Drive or a reverb send.

Slicer Slice Seq uses edges

ENV and RAND advance Slice Seq on each admitted note. CHAOS and NET advance it when their signal crosses zero in the rising direction. The source supplies a trigger count, so changing Amount does not scale the number of steps.

Routes and recall

The pad’s source, destination and amount save with the preset. Envelope levels, random values and the live phases are playback state. Clearing the last route retires that source. Available destinations depend on the pad mode.

For one changing hit at a time, use **RAND**. For slow movement through a sustained note or effect tail, try **CHAOS** or **NET**. Master Sweep has a separate global source selector on page 15.

THIRD CUSTOM RELEASE / +C3

Shared CV utilities

Tools > Rec > CV In. The tab stays labelled Rec; Back returns to Rec controls.

Scale and offset external modulation

CV Scale %	0-400%, default 100%.
CV Offset %	-100 to +100%, default 0%. Added after scaling.
Shared scope	Applies to all eight EXT modulation sources together, not separately per socket. IN1/IN2 audio sources are excluded.

At Scale 200% and Offset -50%, a normalized input value of 0.6 becomes 0.7. The transformed modulation value is limited to -2 to +2; each destination keeps its own range limits. Editing these settings also updates a held voltage without needing the CV to move.

Quantize CV-controlled pitch

Quantize offers Off, Major, Minor, Dorian, Phrygian, Lydian, Mixolydian, Harmonic minor, Major pentatonic, Minor pentatonic, Blues and Whole tone. **Root** selects C through B. Defaults are Off and C.

The native CV-pitch source snaps to the nearest allowed note; ties select the lower note. It is **CV-only**: it does not quantize MIDI notes or the final sum of unrelated pitch modulations. Scale/Offset affect the general EXT modulation values; pitch quantizing operates separately on the native CV-pitch source.

Triggers, saving and reset

Raw CV still reaches the native gate/trigger detector first. Trigger thresholds, rising edges and signal wake remain unchanged. A scale or root edit does not synthesize another trigger. The new Run CV and Reset CV assignments (page 21) also read raw gates, before Scale/Offset.

All four settings save globally through the normal Tools store. **RESET > Tools** restores 100% / 0% / Off / C. RESET > FX leaves them alone. Leaving Rec closes the subpage, so returning to Rec shows its ordinary controls.

For an initial check, keep Scale 100%, Offset 0% and Quantize Off, then change one setting at a time. The default settings preserve native CV behavior.

MASTER CONTROLS +C2 / ROUTING AND RESET +C3

Master processing and RESET

Tools > Comp and Tools > Mstr > Section

Compressor controls [+c2]

Comp page	Threshold -48 to 0 dB; Ratio 1:1 to 20:1; Gain -12 to +24 dB; parallel Mix 0% dry to 100% processed.
Comp env	Tools > Mstr > Section > Comp env: Attack 1-200 ms and Release 10-2000 ms. Defaults: 10 ms and 250 ms.
Defaults	Threshold -4 dB, Ratio 4:1, Gain -4 dB and Mix 100%. The original compressor algorithm remains in use.

Compressor bus [+c3]

Off / Out1/2 / FX1/2 / Both appears under Tools > Main and Mstr > Comp env. These are two views of **the same setting**. FX1/2 means the physical FX output pair. Both uses independent histories/detectors with shared controls. Mono assignments join their output pair before compression.

Master FX [+c2]

Mstr > Section > Master FX offers **Soft, Hard, Fold or Tube** distortion, or Off; Drive adds 0 to +24 dB. Sweep left for low-pass, right for high-pass, center for dry. On Out1/2 the order is distortion, Sweep, then optional compression. These settings save globally; changing compressor buses does not move distortion or Sweep.

RESET with two scopes [+c3]

Tap **Tools > Main > RESET**. Choose Tools or FX, then confirm with **Reset**. Cancel or leave to dismiss.

Tools	Resets global CV, MIDI/clock, recording, routing and master values; saves immediately. Int BPM: 120.0; transport CV/CC and Input FX pad: Off. Keeps **all Display preferences** [+c6].
FX	Resets FX1, FX2 and FX3 scalar settings. Changes the current preset; save the preset to retain the reset. Preserves Tools values.

Both scopes preserve pads, samples, pad parameters, send levels and modulation routes. Neither deletes files or overwrites a saved preset.

EXPANDED MASTER PROCESSING / +C4

Distortion blend and Sweep

Tools > Mstr > Section. Process the main Out1/2 stereo mix.

Blend distortion in parallel

In **Master FX**, select Soft, Hard, Fold or Tube, then set Drive (0 to +24 dB). Switch Section to **Dist Ctrl** for Tone, Dist Mix and Level dB.

Tone	Filters the distorted signal before the blend. Lower it to darken the grit while keeping the clean component bright. Fully open is the default.
Dist Mix	0% clean through 100% distorted, default 100%. The clean and distorted paths are time-aligned.
Level dB	Trims the combined signal from -24 to +12 dB while a shaper is enabled. Default 0 dB. Use it to match loudness after raising Drive.

Modulate the master Sweep

Use the **Sweep** control on the Master FX page to choose the center position. Left is low-pass, center is dry, and right is high-pass. In **Sweep Mod**, choose a source and raise Depth.

Source	Off, EXT1-EXT8 or LFO. EXT uses the corresponding CV jack with the global CV Scale and Offset.
Depth	Signed amount. A negative setting reverses the direction. Defaults are Source Off and Depth 0%.
LFO Rate	One triangle cycle per sixteenth, eighth, quarter or half note, or per 1, 2, 4 or 8 bars. The default is 1 bar. The LFO follows the musical clock independently of GEN Run.

Try a drum loop with Sweep at -40%, Source LFO, Depth +30% and Rate 1 bar. Distortion feeds Sweep, followed by compression when enabled. These master settings save with Tools.

MIDI MONITOR +C2 / RECORDING UPDATES +C5

MIDI monitor and recording

Tools > MIDI > Monitor. Recording workflows continue on pages 17-19.

Recording keeps the pad settings

FX1/FX2 sends [+c2], FX3 and send routes [+c3], and Dry Level/routes [+c5] survive recording. With Input FX enabled, monitor effects follow the recording destination during a take, then return to the saved Follow/fixed choice. **Input FX pad: Off** bypasses pad processing throughout, including during recording [+c8; page 23].

Monitor On / Auto / Off keeps its native meaning. Ordinary input recording writes the input including gain; monitoring effects stay out of the WAV. Use overdub Post+FX or Resam when you want to print processed playback.

Tools > MIDI > Monitor [+c2]

Tap Monitor beside the version number. Capture starts on entry. Seven visible rows show the newest entries from a 32-event history. Each row gives elapsed time, input group, channel and message. **0.350 A 03 CC 74=96**, for example, means CC 74 at value 96 on channel 3 after 350 ms.

Freeze / Resume	Freeze holds the displayed history while capture and playback continue. Encoder 2 scrolls frozen rows. Resume returns to the latest events.
Channel filter	Touch Ch All / Ch 1-16 or use encoder 1. Transport messages appear with every channel setting.
Clear / Back	Clear restarts elapsed time and clears the log. Back stops capture and returns to MIDI settings.
CLK / SENSE	Separate activity lights for MIDI Clock and Active Sensing. Rows show notes, CC, program changes, pressure, pitch bend, transport and reset messages.

A/B are native input groups. B includes the USB queue and one parser path, so a group is not a unique jack identifier. SysEx and system-common messages are discarded before this monitor. Signal Wake is configured separately under Disp.

Recording with a stopped clock [+c7]: Rec starts the clock for the take, then restores its stopped state afterward. An already-running clock continues. Quantization and threshold still apply. Navigation and gain controls also work on ordinary input takes; see pages 17-18 for overdub and Resam.

LIVE RECORDING / +C5

Overdub a loop or performance

Empty destination > INFO twice > Cnf > Dub source / Dub tap

Prepare the destination

Select an empty pad. In **Cnf**, choose **Dub source** and **Dub tap**. Return to Load and Record, choose **Rec Input > In 1/2** and use **Length > Custom** for a take you end manually. Recording creates a new WAV; it does not overwrite the backing sample.

Off	Ordinary input recording, without a pad backing contribution.
Manual	Record stereo input with the live musical pads you play. Already-playing pads enter at their current positions. Nothing starts or stops automatically. Leave the recorder for Pads to trigger, retrigger or stop sounds; MIDI, CV and GEN remain under your control.
Pad 1-16	Rec restarts the chosen source, even if already playing, then stops it when the take ends. Other musical pad taps are ignored during the take. Numbers run left to right, top to bottom: Pad 4 is upper-right, Pad 8 directly below.
Unavailable source	An empty numbered pad or the destination itself shows an explanation and keeps the recorder open. Choose another loaded source in Cnf and retry.

Loop underneath, record for as long as needed

The source uses its normal Sample, Clip, Slicer or Granular playback. Set the source's Loop Mode for a repeating backing, then record a take of any duration using Custom and Stop. Recording quantization and threshold still determine the accepted start.

One pass for eligible Sample sources

A numbered **Sample** source with **Pre**, forward playback, Loop Off or Forward, and no loop crossfade exposes **Dub take**. **Until Stop** follows playback for the take; **One pass** finishes at the first loop wrap or playback end, following live Pitch. Other combinations use recording Length or Stop.

Tap the recording destination on Pads to stop the take. The next INFO opens that destination. To reopen without stopping, select the destination with a left-hand encoder, then press INFO. More controls on page 18.

LIVE RECORDING CONTROLS / +C5

Recording taps, gain and Resam

Keep recording while you play or edit other pads.

What the backing contributes

Pre	Live timing, Pitch, Speed, level, pan and envelope, before Fidelity, Drive, pad filters and Dry Level.
Post pad	Pre plus Fidelity, Drive and the pad filters, including modulation, followed by Dry Level. Dry 0% removes this direct contribution.
Post+FX	Post pad plus shared FX1 delay and FX2/FX3 reverb returns, before physical output routing and master processing.

All three add live stereo input. Shared returns can contain other pads' tails or effects on monitored input. For an isolated wet backing, stop other sources and let tails decay. Post+FX does not print master compression, distortion or Sweep.

Resample the mix

Rec Input > Resam captures the native resampling mix. Dub source/tap do not add a second backing copy in this mode. Play or edit pads while recording, then stop from the recorder or by tapping its destination on Pads.

Adjust level during the take

Either **right-hand encoder** adjusts **Input Gain** or **Resam Gain**, from -96 to +20 dB. Input Gain changes the incoming audio, leaving backing pads and shared returns unchanged. Resam Gain changes the written mix without changing playback volume. Gain is a native global setting.

Leave, return and stop

HOME can return to Pads while the take continues. Select the recording destination with a left-hand encoder and press INFO to restore **Stop, Gain, timers and meters**. Tapping that destination on Pads requests Stop and selects it for the next INFO. Stop still honors recording quantization; normal settings return when the take finishes.

For direct playback of a printed take, restore the destination's **Dry Level to 100%** and lower its sends to avoid applying effects twice. These settings are deliberately preserved through recording.

DRY LEVEL +C5 / LIVE INPUT +C8

Dry to fully wet

Pad > INFO twice > Pos > Dry Level, after FX3 Send

Balance the direct sound and three parallel effects

Dry Level trims each pad's direct signal from **100%** (default) down to **0%**. The FX1, FX2 and FX3 sends receive the signal before this attenuation, so their effect contributions remain available. Settings save with the pad; older presets start at 100%.

Ordinary blend	Keep some Dry Level and raise the desired sends. Use each effect's return level to set the audible effect amount.
Wet only	Set Dry Level to 0%, then raise one or more sends. FX1 delay and both reverbs can contribute while the direct pad is silent.
One effect only	Set Dry Level to 0%, raise that effect's send and lower the other two. Shared effects may still contain tails from other pads.
Modulation	Select Dry Level and press INFO. Three slots share the existing total of 12 routes per pad. The base level plus modulation clamps to 0-100%; double-tap resets the base to 100% without clearing routes.

Parallel sends, not three independent crossfaders

There is one direct path per pad. Dry Level controls it; three independent send amounts and the effect returns control the wet paths. For a crossfade, modulate Dry downward while modulating the desired send upward. Dry modulation follows the same pad-mix and voice-source rules as the sends on page 11.

Recording and overdub

Recording preserves Dry Level and its modulation. Pre overdub is captured before Dry attenuation; Post pad and Post+FX follow it. At Dry 0%, Post+FX can print wet-only backing. Raw input capture stays unchanged. **[+c8]** When Input FX is enabled, its pad's Dry Level trims live monitoring while sends stay independent. **Input FX pad: Off** bypasses this trim and pad processing; see page 23.

When an effect is routed to another physical output, listen to that output to hear the wet signal. Routing controls are on pages 9-10. Dry Level does not change those assignments.

CLOCK AND TRANSPORT / +C6

Clock and internal tempo

Tools > Main > Int BPM / Clock

Tools > Main

Upper left	**Int BPM**: 40.0-300.0 BPM, default 120.0. Slow turns stay fine; fast turns accelerate. Pause or reverse for fine control again. Edits save and keep phase.
Lower left	**Clock**: tap to open transport controls. The tile reports the observed clock source and Run/Stop state.
Right column	Existing **MasterVol** and **Compressor**. Brightness and Flip UI have moved to Disp; see page 2.

Hold Int BPM for three seconds to restore and save **120.0 BPM**. Release, move outside, turn its encoder or leave the page to cancel. Only internal tempo resets; transport and active external clock remain unchanged.

Clock header: Run / Stop / Reset

Run · green	Resumes musical time from its held position.
Stop · red	Holds musical time at its current position.
Reset · blue	Returns musical position to the beginning; keeps its running or stopped state. Shown in the native cyan-blue accent.

HOME returns to the main Menu; **Back** returns to Tools Main. The footer shows source, actual BPM and state. Sounding Samples and tails can continue after Stop; native quantized launches can restart transport. Pitch and Sample Speed are independent of BPM.

Keep the last external tempo

External clock takes priority. CV-to-valid-MIDI fallback keeps native priority. When the final external source times out, internal clock continues at its last tempo and phase. **Losing clock pulses does not issue Stop.**

Int BPM shows the inherited tempo, but it is **temporary**. Edits start there and save your choice. Reboot uses the last explicitly saved tempo. Tools Reset restores 120.0 BPM.

Counter double-tap only renumbers the next bar. **Clock > Reset** resets musical time. Synced playback retains its normal clock-dependent behavior.

CLOCK INPUTS +C6 / RECORDING FIX +C7

CV and MIDI transport

Tools > Main > Clock. Assignments save globally; all CV and CC controls default Off.

Clock grid

Run CV	Left, upper: Off / EXT1-EXT4. A rising gate resumes; a falling gate stops.
Reset CV	Left, lower: Off / EXT1-EXT4. A rising gate resets; a falling gate has no reset action.
Reset CC	Right, upper: Off / CC 0-119. A selected CC crossing from 0-63 to 64-127 resets once. Send below 64 to rearm; repeated high values do not reset again.
Reset Ch	Right, lower: MIDI channel 1-16 for Reset CC; default 1.

Assign both CV controls to one input for **reset-and-run on the rise, Stop on the fall**. Enabling an already-high gate creates no edge: cycle it afterward. Transport reads raw CV before Scale/Offset. Ordinary CV and CC routing stays active; use an otherwise unused CC if reset should have no other effect.

Standard MIDI transport stays compatible

Start / FA	Resets to the beginning and runs.
Continue / FB	Resumes the held musical position.
Stop / FC	Holds position without resetting.

The clock-port selector filters Start/Continue/Stop. Reset CC listens to either MIDI input using its own Reset Ch/CC. System Reset (FF) is not repurposed. Reset CC works while running or stopped without replacing standard transport.

GEN and recording

Stop halts GEN scheduling, releases its held gates, cancels unfinished rolls and restores temporary Cut settings. Continue keeps pattern state and resumes future steps without replaying a partial step. Reset prepares the beginning; a stopped GEN stays silent until Run. Its saved GEN Run switch remains independent.

Rec [+c7] temporarily starts a stopped clock and restores Stop after the take; an already-running clock keeps running. **Rec busy:** transport commands are ignored while armed, recording or completing the final overdub FX write. Stop the take with the recorder controls first. Rejected transport commands are not replayed afterward.

PRESET WORKFLOW / +C6

Save and recall pad mutes

Use the existing Mixer mute controls; there is no new mute menu.

Keep a starting mix with a preset

1. Set the musical pads to muted or unmuted in **Mixer**. A mute edit now marks the preset as changed. It does not automatically write the SD card.

2. Save the preset normally. This stores the mute state of each of the sixteen musical pads alongside its other settings.

3. Reload the preset to restore the saved mix. Both the native mute indication and the playback mute return. Saving, rebooting and loading the same preset recalls those states too.

What carries over

Older presets	Presets saved before +c6 have no stored mute field. They load with all musical pads **unmuted**, regardless of the previous preset's mute states.
Copy a pad	The copy carries the source pad's mute setting.
Clear or replace	Clearing/unloading a pad, replacing its WAV or recording a new take clears mute, following the existing reset behavior. Save again to remember the result.
Tools / FX reset	Neither reset scope changes pad parameters or saved pad mutes. Tools Reset also preserves all Display preferences; see pages 2 and 14.

A useful check when a pad is silent

After loading a preset, check its Mixer mute as well as its volume, output and Dry Level/send settings. A saved-muted pad is deliberately recalled silent; this does not mean its sample or assignment is missing.

On **Tools > Main** and **Rec > CV In**, the second gray header box shows SD-card free space. Use **RESET** for Tools/FX defaults and **Clock > Reset** for transport.

All feedback is welcome. Join Galagzee Audio on Discord and visit **#bitbox2-custom-firmware** for questions, ideas, reports of problems or experiences with your setup.

LIVE INPUT PROCESSING / +C8

Using Bitbox Mk2 as an effect box

Use existing pad controls without loading a sample or starting a recording.

Set up the input and sound

Connect audio to **IN1/2**. Select an empty pad, press **INFO** and set **Rec Input: In1/2**, **RecMon: On** and Gain. Scroll to **Input FX pad**: **Off (default) > Follow > Pad 1 > ... > Pad 16**. Turn right from Off to enable processing; no recording is needed.

Off bypasses pad Filter/Tone, Dry Level and sends, even during recording. Native Gain, RecMon and Mon Output still work. Fresh settings and Tools Reset use Off; updating preserves existing saved Follow/fixed choices. Select Off manually to disable one.

With Follow or a fixed pad selected, press INFO again. **Main:** Filter / Resonance. **Tone:** Fidelity, Drive, Filter 2 / Res 2, Slope 1. **Pos:** FX1/2/3 Send, Dry Level. Configure the shared effects too. Processing feeds direct audio and sends; Dry 0% leaves only effects.

Fixed pad or quick Follow presets

There is **one global selector**. Fixed **Pad 1-16** keeps that pad's controls while browsing. **Follow** uses the selected pad: prepare empty pads with different sounds, then tap between them. Their settings are not combined. Save the preset to retain pad controls; the Input FX choice saves globally.

Choose the outputs; switch the audible combination

Recorder > Mon Output routes direct input. **Tools > Rec > FX Out** routes shared FX1/FX2 returns; **FX3 Out** is separate. For everything on one pair, choose it in all three places. **Cnf > Output** routes sample playback, not input. Pad taps do not recall these routes or shared FX settings.

Example: Mon Output **Out1/2**; FX Out and FX3 Out **FX1/2**. Select **Follow**:

Pad A: direct	Dry 100%, all sends 0: input on **OUT1/2**.
Pad B: wet	Dry 0%, sends raised: effects on **FX OUT1/2**.
Pad C: blend	Dry and sends raised: **both output pairs**. Each pad can have different Filter/Tone values.

Tails may continue across pad changes or Off. **FX1 > Purge** clears delay, not reverb. Input Filter/Tone uses **set values only**; Normal/High Q leave input unchanged. Send/Dry modulation works; ordinary input recording stays dry. With Input FX enabled, a take uses its destination's monitor controls, then restores the saved choice; Off always bypasses them.

AUDIO DEADLINE PRESSURE / +C9

Audio load meter

Top toolbar readout; saved visibility under Tools > Disp

Read the percentage

The top toolbar shows a **smoothed percentage** of the audio-processing time budget. Compare readings as you add voices, Speed or effects. Smoothing and a small deadband keep ordinary changes readable.

Normal percentage	A smoothed view of audio deadline pressure, including time waiting for an audio job to run and its processing through output completion.
Red 100%+	A measured budget excursion or audio-queue problem. The warning bypasses smoothing and stays visible for **one second** after the most recent event.
--%	No fresh valid measurement is available. It is not a zero-load reading.

A red warning **can occur without audible crackling**. This is a practical audio-load indicator, not a measurement of whole-device CPU utilization or a guarantee that every audible glitch will be detected.

Show or hide it

Open **Tools > Disp**. The header button between **Gfx** and the version reads **CPU On** or **CPU Off**. Tap to switch. **On is the default**, including when loading settings from earlier firmware.

The choice saves automatically, survives a restart, and is preserved by both Tools and FX reset. **Off** removes the percentage and its UI updates; lightweight audio timing collection continues so switching remains safe during playback. Re-enabling resumes the smoothed reading without replaying expired warnings.

Status messages and screen saver

Native status text and progress displays have priority in the toolbar. GEN can use the freed width when the meter is hidden. Meter updates do not wake the screen saver.

Using the warning

If the warning is frequent or you hear unwanted artefacts, compare with fewer active voices or effects. A short red event by itself does not mean the sound has failed. The meter helps judge a particular patch under load; it does not add more processing capacity.

NINTH CUSTOM RELEASE / +C9 AUDIO LOAD METER

Install, recover and credits

Bitbox MkII only. The patcher always takes official firmware 2.3.4 as input.

Patch locally and install

1. **Back up the entire microSD card**, including projects, samples and presets. Download official 2.3.4 and extract the ZIP. At the browser patcher, select the original BITBOXMK2.bin, even when upgrading from an earlier custom release. Patching runs in your browser; the firmware is not uploaded. Input and output SHA-256 checks guard the patch.

2. Put the patched file in the card root, named exactly **BITBOXMK2.bin**. Remove duplicate-download suffixes such as (1). Power off, insert the card, then hold **INFO** while powering on. Release INFO when the update message appears.

3. Allow about 30 seconds. Keep power on and wait for automatic reboot. Check **2.3.4+c9** at Menu's lower-right or Tools' top-right. See the official MkII manual for the complete flashing procedure.

Official input: 674,896 bytes. SHA-256:
35ab7995ad0eca8484f3b28bc76389f10b2037dbb21a4a394e950d9329073e86

Customized +c9 output (786,412 bytes). SHA-256:
64d8afce76cae837bb0ddc99161958eb69d1e681700c04f305dd350dbf5cd791

Return to official firmware

Replace the patched file on the card with the original unmodified **BITBOXMK2.bin** from the official ZIP. Repeat the INFO-held power-on update and wait for reboot. Confirm **2.3.4** in Menu. Keep backups of presets made before upgrading.

Reference and acknowledgment

First release (+c) video overview. The video demonstrates the original features; later-release controls are covered in this guide.

1010music created the firmware and hardware. vwal integrated the custom features and patcher. GEN, the modulation sources and several effects draw on bitbox-mk2-semi. Upstream work includes Portkit, Justin Joe’s blackbox-mod, Emilie Gillet’s stmlib and OpenTracker-derived GEN procedures. See third-party notices for attribution and license details.

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